

JAVA PROGRAMMING LAB

Experiment No: 05

Date: 21/09/26

Title: Implementation of Set Interface Methods in Java Using HashSet

1. Objective / Aim

To write and execute a Java program using the Set interface and HashSet implementation to demonstrate commonly used Set methods such as add(), remove(), contains(), size(), isEmpty(), and clear(), and to observe that duplicate elements are not stored in a Set.

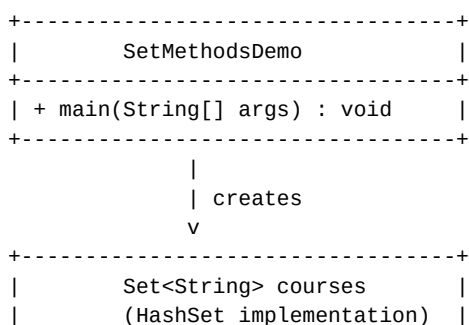
2. System Requirements

Hardware Requirements	Software Requirements
Computer / Laptop	Operating System: Windows / Linux / macOS
Minimum 4 GB RAM	JDK 17 or JDK 21
Minimum 10 GB free storage	Java compiler (javac) and Java Runtime (java)
Keyboard and Mouse	IDE: Eclipse / IntelliJ IDEA / NetBeans / VS Code (optional)

3. Algorithm / Logic Description

1. Start the program.
2. Import HashSet and Set from java.util.
3. Create a Set named courses using new HashSet<>();
4. Add Java, Python, and C to the set using add(). Add Java again to demonstrate that duplicate values are not added.
5. Remove C using remove().
6. Check whether Python exists using contains().
7. Find the number of elements using size().
8. Check whether the set is empty using isEmpty().
9. Remove all elements using clear().
10. Check isEmpty() again after clear().
11. Display the results and stop the program.

4. UML Class Diagram



```
+-----+
| add()           |
| remove()        |
| contains()       |
| size()           |
| isEmpty()        |
| clear()          |
+-----+
```

Note: Set is an interface, while HashSet is the concrete implementation used to store the course names.

5. Source Code

```
import java.util.HashSet;
import java.util.Set;

class SetMethodsDemo {

    public static void main(String[] args) {

        // Creating a Set
        Set<String> courses = new HashSet<>();

        // add(E e)
        courses.add("Java");
        courses.add("Python");
        courses.add("C");
        courses.add("Java");    // Duplicate value is not added

        System.out.println("After add(): " + courses);

        // remove(Object o)
        courses.remove("C");
        System.out.println("After remove(): " + courses);

        // contains(Object o)
        System.out.println("Contains Python? " + courses.contains("Python"));

        // size()
        System.out.println("Set size: " + courses.size());

        // isEmpty()
        System.out.println("Is set empty? " + courses.isEmpty());

        // clear()
        courses.clear();
        System.out.println("After clear(): " + courses);

        // Checking again
        System.out.println("Is set empty after clear()? " + courses.isEmpty());
    }
}
```

6. Completion and Execution Commands

1. Save the program as:
SetMethodsDemo.java
2. Compile:
javac SetMethodsDemo.java
3. Execute:
java SetMethodsDemo

7. Output

```
After add(): [Java, Python, C]
After remove(): [Java, Python]
Contains Python? true
Set size: 2
```

```
Is set empty? false
After clear(): []
Is set empty after clear()? true
```

The exact order of elements printed by HashSet may vary because HashSet does not guarantee insertion order. The important results are that the duplicate Java is not added, C is removed, Python is present, the size becomes 2, and clear() makes the set empty.

8. Result / Completion

The program was successfully written and executed to demonstrate the basic methods of the Set interface using HashSet. The behavior of duplicate elements, removal, searching, size checking, empty checking, and clearing was verified.